COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S4
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ACTIVITY NO. 1 REVIEW OF C++ PROGRAMMING

1. PROCEDURE OUTPUT

ILO A: Create Code That Follows the Basic C++ Code Structure

Create a function that will take two numbers and display the sum.

ilo_a.cpp ilo_c.cpp

HOA1 > ilo_a.cpp > ...

```cpp
#include <iostream>

//Create a function that will take two numbers and display the sum.
//prototyping:
int sumOfTwoNum(int fNum, int sNum);

int main() {
    std::cout << "The sum of the two numbers is: " << sumOfTwoNum(6,7)<< std::endl;
}

//define
int sumOfTwoNum(int fNum, int sNum) {
    return fNum + sNum;
}
```

```
The sum of the two numbers is: 13
PS D:\Downloads (HDD)\CPE010>
```

Create a function that will return whether variable A is greater than variable B.

ilo_a.cpp ● ilo_b.cpp ilo_c.cpp

HOA1 > ilo_a.cpp > ...

```cpp
#include <iostream>

//Create a function that will return whether variable A is greater than variable B.
//prototyping:
int greaterVar(int varA, int varB);

int main() {
    if (greaterVar(0,7)){
        std::cout << "Variable A is greater than Variable B\n";
    } else {
        std::cout << "Variable A is NOT greater than Variable B\n";
    }
}

//define
int greaterVar(int varA, int varB) {
    return varA > varB;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Variable A is NOT greater than Variable B
PS D:\Downloads (HDD)\CPE010>
```

ilo_a.cpp | ilo_b.cpp | ilo_c.cpp

HOA1 > ilo_a.cpp > main()

```cpp
#include <iostream>

//Create a function that will return whether variable A is greater than variable B.
//prototyping:
int greaterVar(int varA, int varB);

int main() {
    if (greaterVar(21,19)){
        std::cout << "Variable A is greater than Variable B\n";
    } else {
        std::cout << "Variable A is NOT greater than Variable B\n";
    }
}

//define
int greaterVar(int varA, int varB) {
    return varA > varB;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
Variable A is greater than Variable B
PS D:\Downloads (HDD)\CPE010>
```

Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.

ilo_a.cpp | ilo_a2.cpp | PartA.cpp | ilo_b.cpp | ilo_a3.cpp | ilo_c.cpp

HOA1 > ilo_a3.cpp > ...

```cpp
#include <iostream>

// Create a function that will take two Boolean values and display the result of all logical operations then
// return true if it was a success.
bool logicOperation(bool a, bool b) {
    std::cout << std::boolalpha; //displays Boolean values

    std::cout << "Boolean Logical Operators" << std::endl;
    std::cout << "If a = " << a << " and b = " << b << ", then..." << std::endl;

    std::cout << "AND (a && b)   : " << (a && b) << std::endl;
    std::cout << "OR  (a || b)   : " << (a || b) << std::endl;
    std::cout << "NOT (!a)   : " << (!a) << std::endl;
    std::cout << "NOT (!b)   : " << (!b) << std::endl;

    return true;
}

int main() {
    return logicOperation(true, false);
}
```

```
Boolean Logical Operators
If a = true and b = false, then...
AND (a && b)    : false
OR  (a || b)    : true
NOT (!a)        : false
NOT (!b)        : true
PS D:\Downloads (HDD)\CPE010>
```

ILO B: Implement Appropriate Class Definition and Instances Based on Given Requirements

Input/Output Code

HOA1 > ilo_b.cpp > main()

```cpp
#include <iostream>

class Triangle {
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngle(double A, double B, double C);
    bool validateTriangle(); //remove const
};

Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}

void Triangle::setAngle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A + B + C;
}
```

```cpp

// Used AI on how to validate Triange
bool Triangle::validateTriangle() {
    return (totalAngle == 180.0 && angleA > 0 && angleB > 0 && angleC > 0);
}

int main() {
    //driver code
    Triangle set1(40, 30, 110);

    if (set1.validateTriangle()) {
        std::cout << "The shape is a valid triangle. \n";
    } else {
        std::cout << "The shape is not a valid triangle. \n";
    }

    return 0;
}
```

```cpp
int main() {
    //driver code
    Triangle set1(0, 0, 0);

```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

PS D:\Downloads (HDD)\CPE010> ^C
PS D:\Downloads (HDD)\CPE010>
PS D:\Downloads (HDD)\CPE010>  & 'c:\Users\
-MIEngine-In-fpxnybfk.jg0' '--stdout=Micros
fs' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe'
The shape is not a valid triangle.
```

```
32  int main() {
33      //driver code
34      Triangle set1(40, 30, 110);
35
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

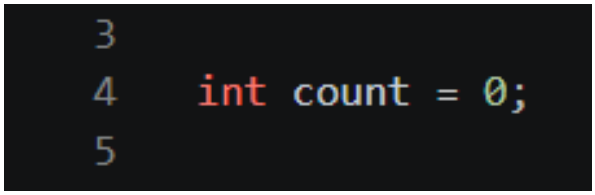
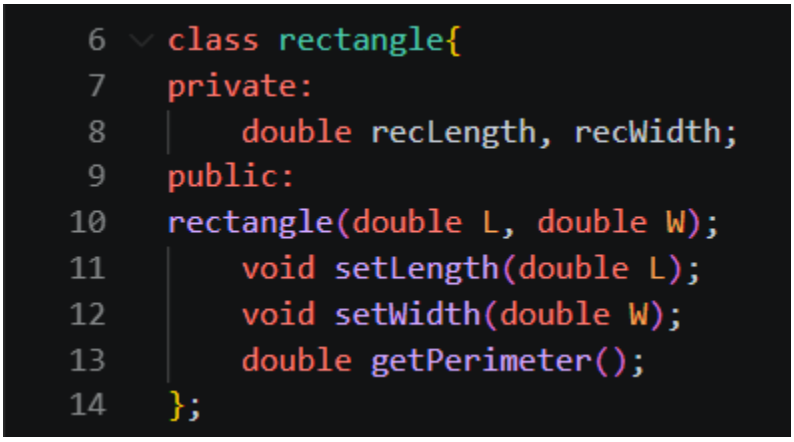
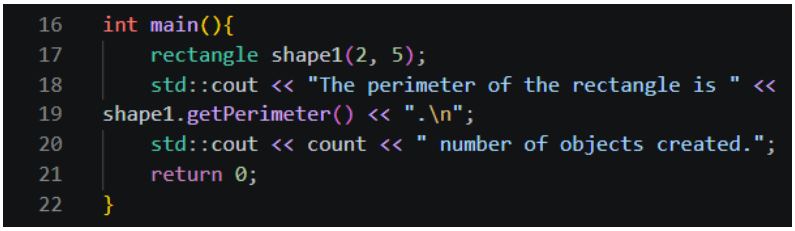
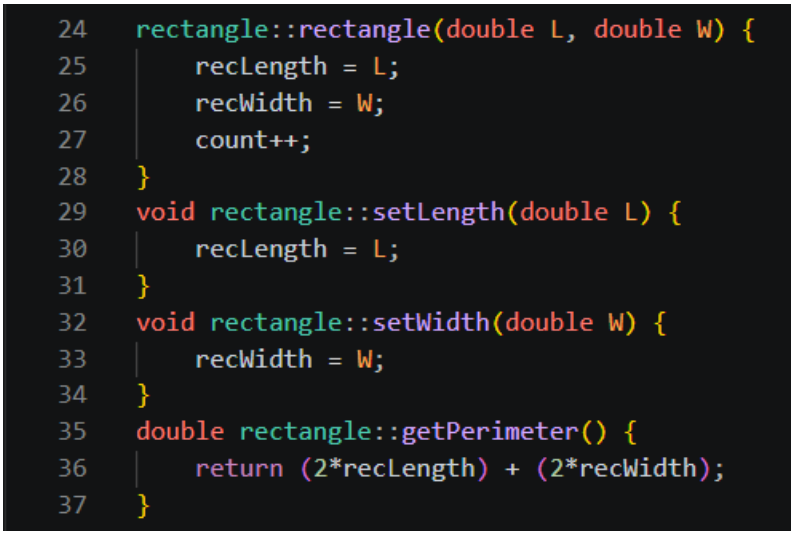
PS D:\Downloads (HDD)\CPE010> ^C
PS D:\Downloads (HDD)\CPE010>
PS D:\Downloads (HDD)\CPE010>  & 'c:\Users
-MIEngine-In-2wbx4ymy.bgm' '--stdout=Micro
u3' '--dbgExe=C:\msys64\ucrt64\bin\gdb.exe
The shape is a valid triangle.
PS D:\Downloads (HDD)\CPE010>
```

Code Analysis

I've tried to research if "const bool" is appropriate to use on the existing code, then Google said its possible, but not for this code as my assignment operator for calculating if my triangle is valid, would not function without it; so to fix this, I deleted const keyword, for the code to reduce its error. Next is to figure out how to fix validation of the triangle; if "TotalTriangle" accepts all sum values of AngleA, AngleB and AngleC, then all numbers will count as valid triangle, which should not happen and no error would occur. So I asked for AI to help me on what I should do, then the answer shown to me was quite simple yet I couldn't figure it out, all I had to do was to use an Assignment Operator to note that all angles can't be lesser than 0 and should all sum exactly to 180 degrees.

C++ Structure Code for Answer

Header File Declaration Section	ilo_a.cpp PartA.cpp HOA1 > PartA.cpp > ... 1 #include<iostream> 2 using namespace std; 3

Global Declaration Section	
Class Declaration and Method Definition Section	
Main Function	
Method Definition	

```
3
4    int count = 0;
5
```

```
6   class rectangle{
7   private:
8       double recLength, recWidth;
9   public:
10  rectangle(double L, double W);
11      void setLength(double L);
12      void setWidth(double W);
13      double getPerimeter();
14  };
```

```
16  int main(){
17      rectangle shape1(2, 5);
18      std::cout << "The perimeter of the rectangle is " <<
19  shape1.getPerimeter() << ".\n";
20      std::cout << count << " number of objects created.";
21      return 0;
22  }
```

```
24  rectangle::rectangle(double L, double W) {
25      recLength = L;
26      recWidth = W;
27      count++;
28  }
29  void rectangle::setLength(double L) {
30      recLength = L;
31  }
32  void rectangle::setWidth(double W) {
33      recWidth = W;
34  }
35  double rectangle::getPerimeter() {
36      return (2*recLength) + (2*recWidth);
37  }
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C: Solve Different Problems using the C++ Programming Language

Create a C++ program to swap the two numbers in different variables.

ilo_a.cpp | ilo_a2.cpp | PartA.cpp | ilo_b.cpp

HOA1 > ilo_c.cpp > ...

```cpp
#include <iostream>
using namespace std;

int main() {
    int num1, num2, temp;

    cout << "Input 1st number: ";
    cin >> num1;
    cout << "Input 2nd number: ";
    cin >> num2;

    cout << "\nOriginal Numbers before swapping:" << endl;
    cout << "First number = " << num1 << endl;
    cout << "Second number = " << num2 << endl;

    // swap
    temp = num1;
    num1 = num2;
    num2 = temp;

    cout << "\nAfter swapping 1st and 2nd number:" << endl;
    cout << "First number = " << num1 << endl;
    cout << "Second number = " << num2 << endl;

    return 0;
}
```

```
Input 1st number: 12345
Input 2nd number: 00009

Original Numbers before swapping:
First number = 12345
Second number = 9

After swapping 1st and 2nd number:
First number = 9
Second number = 12345
PS D:\Downloads (HDD)\CPE010>
```

Create a C++ program that has a function to convert temperature in Kelvin to Fahrenheit.

ilo_a.cpp | ilo_a2.cpp | PartA.cpp | ilo_b.cpp | ilo_a3.cpp

HOA1 > ilo_c2.cpp > ...

```cpp
#include <iostream>
#include <iomanip>

float kelToFar(float kelvin);

int main() {
    float kelvin_num = 0;

    std::cout << "Kelvin to Fahrenheit Conversion" << std::endl;
    std::cout << "Input temperature in Kelvin: ";
    std::cin >> kelvin_num;

    float fahrenheit = kelToFar(kelvin_num);

    std::cout << std::fixed << std::setprecision(2);
    std::cout << "Fahrenheit: " << fahrenheit << " °F" << std::endl;

    return 0;
}

float kelToFar(float kelvin) {
    return ((kelvin - 273) * 1.8) + 32;
}
```

```
Kelvin to Fahrenheit Conversion
Input temperature in Kelvin: 100
Fahrenheit: -279.67 °F
PS D:\Downloads (HDD)\CPE010>
```

Create a C++ program that has a function that will calculate the distance between two points.

HOA1 > ilo_c3.cpp > ...

```cpp
#include <iostream>
#include <cmath> // importing cmath library for sqrt

float coordDistance(float x1, float y1, float x2, float y2);

int main() {
    float x1, x2, y1, y2;

    std::cout << "Input x-coordinates for 1st Point: ";
    std::cin >> x1;
    std::cout << "Input y-coordinates for 1st Point: ";
    std::cin >> y1;

    std::cout << "Input x-coordinates for 2nd Point: ";
    std::cin >> x2;
    std::cout << "Input y-coordinates for 2nd Point: ";
    std::cin >> y2;

    float distance = coordDistance(x1, y1, x2, y2);

    std::cout << "Distance between two points is: " << distance << std::endl;

    return 0;
}

float coordDistance(float x1, float y1, float x2, float y2) {
    return std::sqrt((x2 - x1) * (x2 - x1) + (y2 - y1) * (y2 - y1));
}
```

```
Input x-coordinates for 1st Point: 10
Input y-coordinates for 1st Point: 10
Input x-coordinates for 2nd Point: 20
Input y-coordinates for 2nd Point: 20
Distance between two points is: 14.1421
PS D:\Downloads (HDD)\CPE010>
```

Conclusion

For my conclusion for this activity is I've relearn on how to use the basics of C++ on ILO A with creating a function that will take two numbers and display the sum, create a function that will return whether variable A is greater than variable B and creating a function that will take two Boolean values and display the result of all logical operations then return true if it was a success. On ILO B, I looked into whether "const bool" could be used on the current code, and Google said it could, but not for this code because my assignment operator, which determines whether my triangle is valid, wouldn't work without it; In order to reduce the error in the code, I removed the const keyword. The next step is to figure out how to validate the triangle. If "TotalTriangle" accepts all sum values of AngleA, AngleB, and AngleC, then all numbers will count as valid triangles, which shouldn't happen and there wouldn't be an error. I asked AI to advise me on what to do, and although the answer was fairly straightforward, I was unable to understand it. All I needed to do was use an Assignment Operator to note that all angles must add up to 180 degrees and cannot be less than 0. I've also solved how to create a C++ program to swap the two numbers in different variables, create a C++ program that has a function that will calculate the distance between two points and create a C++ program that has a function that will calculate the distance between two points.

I apologize for taking too long to submit my activity as I was not committed as I am now to pass it in due time, not realizing this kind of assessment will not stop being assigned to us. I will now change my approach on creating my work and passing it in due time to reduce my workload and move on to different activities to maximize my time. Reality has caught up to me, so I have to carry out my duty and responsibility as a 2nd year student and hope I won't repeat this mistake again or I will surely won't catch up to my blockmates in progressing. I should focus more on specific errors that I couldn't solve on my own and analyze what I should do in the future to continue advancing to difficult lessons.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **QuillBot, Google AI Overview**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Grammar Correction, Code Assistance, Explanation of Concept**

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Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Assistance on how to identify the errors of my code and to understand my thoughts better in written form**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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